

Response to Reviewer’s comments

Reviewer #1 Comments and Response:

We sincerely thank the reviewer for the constructive comments and valuable suggestions. All comments have been carefully considered, and the manuscript has been revised accordingly. Detailed responses are provided below.

Comment No.	Reviewer Comment	Response	Location in Revised Manuscript
1	Please provide stronger evidence demonstrating the advantage of the proposed model over the Linear Regression baseline.	We thank the reviewer for this suggestion. Additional robustness analyses were performed using five random seeds, state-wise evaluation across 10 Indian states, and Diebold–Mariano significance testing. The revised manuscript now explicitly states that the proposed Hybrid Residual model achieves near-parity with Linear Regression (RMSE 0.007187 ± 0.001904 vs. LR 0.006333 ; $R^2 = 0.984295 \pm 0.009279$ vs. LR 0.988452) rather than consistent superiority. The contribution has been reframed as residual decomposition that preserves the strong performance of analytical baselines while modeling localized nonlinear residual behavior.	Abstract, Section VIII-H, Conclusion
2	No statistical evidence for the one claimed benefit. Report how many test points satisfy $y \geq 0.20$, provide confidence intervals or multiple-seed variance, and run a significance test (e.g., Diebold-Mariano) for the LR-vs-proposed difference. A 0.0073 vs 0.0073 global tie and a 6% peak difference without uncertainty quantification cannot support the stated conclusions.	Addressed comprehensively: (1) Peak subset quantified: 114 of 1266 test points satisfy $y \geq 0.20$ (9.0%). (2) Five-seed evaluation added with mean and std for all metrics. (3) DM test on full-test and peak-test subsets: mean DM statistic = -0.39 , $p = 0.20$ —not significant at $\alpha = 0.05$. Only 1 of 5 seeds achieves significance. (4) Conclusion revised to state near-parity.	Section VI-C, VI-D, VI-E, Table 8

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3	The “optimized oracle export” results are unexplained and implausible as reported. Table 8 shows an “optimized oracle export” RMSE of 0.000170. Define precisely how these 15-minute streams are produced, what is being optimized against what, and confirm there is no use of test-period targets. If this cannot be justified, remove it.	We agree this was confusing and potentially misleading. The “optimized oracle export” table has been completely removed from the revised manuscript. It referred to an internal intermediate processing step that was incorrectly reported as a forecasting result. All evaluation now uses only the standard model outputs with clearly documented metrics.	Former Table 8 removed; Section VI reorganized
4	The blockchain layer is described but not evaluated. Either provide a concrete prototype evaluation (even on a local EVM test-net with measured gas/latency) or reframe the blockchain material as proposed architecture and temper the claims accordingly.	We have implemented and evaluated a complete three-contract Solidity prototype on a local Hardhat Ethereum node (Solidity 0.8.24). A 12-module testing suite (62 test cases, all passing) validates: gas economics (anchorForecast 249,222 gas), pipeline latency (2.47 ms), throughput (709 TPS), storage compression (2.24:1), block finality at multiple mining intervals, and PGS gating correctness. Transaction confirmation time corresponds to local Hardhat block confirmation under automine settings; production-network confirmation times were not evaluated and remain future work.	Section VII-G (new), Tables XII–XVII (new)

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5	The Predictive Green Score is not defined. The intro describes a 0–100 “green score” used as a smart-contract threshold; Section III uses a 7000/10000 on-chain integer threshold. How the confidence/green score is computed from the model output is never specified. Give the exact formula and show its calibration.	PGS now explicitly defined: $PGS_t = \text{round}(10000 \cdot \text{clip}(\hat{y}_t, 0, 1))$. Scheduling gate: $E_t = \mathbb{K}[\hat{y}_t \geq 0.15] = \mathbb{K}[PGS_t \geq 1500]$. This is a deterministic forecast-strength score, not a calibrated confidence score. Previous scale inconsistency removed. Calibrated uncertainty-aware PGS design listed as future work.	Section I, Section VII-B, Section IX
6	Scheduling benefit is a property of the threshold, not of the model. Compare scheduling outcomes under LR vs the proposed model vs perfect-foresight, on the same gating rule. Without that comparison the 68.26% figure does not isolate the value of your method.	New comparison: LR: 280 eligible hours, 25.81% renewable share; Hybrid Residual: 280 hours, 25.81% (identical); Perfect foresight: 290 hours, 25.74%. The revised text explicitly states the operational gain is attributed to the thresholded selection rule, not to a forecast-quality advantage of the Hybrid Residual model over LR.	Section VII-C, Table IX
7	Model naming is inconsistent throughout: “residual decomposition-based model,” “Hybrid Error-Correction,” “Hybrid (LR+BiLSTM)” (figure legends), and “Proposed (residual).” Pick one name and use it everywhere, including figures.	Standardized to “Hybrid Residual” throughout the entire manuscript—all tables, figure captions, running text, and axis labels. All previous variants unified.	Throughout entire manuscript

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8	Table inconsistencies to reconcile: the standalone BiLSTM appears as RMSE 0.0256 in Tables 6 and 9 but EA-BiLSTM v1 is 0.0210 in Tables 5/10—clarify whether these are the same model. Best validation loss precision discrepancy. Validation MAE not in table.	Reconciled: (1) Standalone BiLSTM (full-target) and EA-BiLSTM v1 (energy-aware) clearly labeled as distinct models. (2) Validation loss precision unified to 4 significant figures. (3) Validation MAE now appears consistently where referenced.	Tables V, VI, IX, X; Section V-B text
9	The introduction describes the RNN as operating over “minutes to hours” and “small sequences of hourly values (typically around 10–15 minutes),” which is internally incoherent and contradicts the 48-hour lookback actually used.	Fixed. Introduction now correctly states 48-hour lookback window with hourly temporal resolution. All incorrect references to “minutes” removed.	Section I, Section IV-A
10	The reported near-zero training residual mean (“approximately 0.000000”) is a mathematical property of OLS residuals, not an empirical finding; presenting it as evidence is not meaningful.	Agreed. The statement has been removed. We no longer present the zero-mean residual property as an empirical finding.	Former Section V text removed

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11	Heading typo: “REASEARCH GAPS” (Section II-F). Several capitalization issues persist throughout (lstm/LSTM, bilstm/BiLSTM, “SEQUENCE MODELS,” “Attention,” “MODELS” mid-sentence).	All spelling and capitalization issues corrected throughout, including the heading typo and consistent technical term usage.	Throughout manuscript

Reviewer #2 Comments and Response:

We sincerely thank the reviewer for the constructive comments and valuable suggestions. All comments have been carefully considered, and the manuscript has been revised accordingly. Detailed responses are provided below.

Comment No.	Reviewer Comment	Response	Location in Revised Manuscript
1	Please clarify the exact calculation of the Predictive Green Score, including how forecast confidence is converted to the 0–100 or 0–10000 scale used by the smart contract.	The PGS formula is now explicitly defined: $PGS_t = \text{round}(10000 \cdot \text{clip}(\hat{y}_t, 0, 1))$ with threshold at 1500 (corresponding to forecast ≥ 0.15). Previous scale inconsistency resolved. We clarify this is a deterministic forecast-strength score, not a calibrated confidence metric.	Section I, Section VII-B
2	The proposed model and Linear Regression have almost identical global RMSE, MAE, and R^2 values. Please state more clearly that the contribution is mainly improved peak-period and nonlinear-region behavior, not large improvement in global metrics.	Revised manuscript clearly states that near-parity is expected given the strongly deterministic dataset structure. Contribution reframed as interpretability, failure-mode analysis, and deployable decomposition. Five-seed DM testing provides rigorous statistical evidence.	Abstract, Section VI-A, Section VIII-A, Conclusion

Comment No.	Reviewer Comment	Response	Location in Revised Manuscript
3	Please reduce repeated explanations in Sections VI and VII. Some paragraphs about peak-period RMSE and scheduling importance are repeated and can be shortened.	Redundant explanations consolidated. Peak-period behavior merged into single subsection (Section VI-C). Scheduling interpretation appears once in Section VII-C without repetition.	Sections VI, VII (consolidated)
4	Please provide more detail about the blockchain prototype, including whether gas cost, latency, and transaction confirmation time were measured or only discussed conceptually.	A 12-module testing suite (62 test cases) developed and executed. All gas costs, pipeline latency, block finality, throughput, and storage metrics are now empirically measured. Key results: 709 TPS, 2.47 ms latency, 2.24:1 compression. Transaction confirmation time corresponds to local Hardhat block confirmation under automine settings; production-network confirmation times were not evaluated and remain future work.	Section VII-G (new), Tables XII–XVII (new)

Reviewer #3 Comments and Response:

We sincerely thank the reviewer for the constructive comments and valuable suggestions. All comments have been carefully considered, and the manuscript has been revised accordingly. Detailed responses are provided below.

Comment No.	Reviewer Comment	Response	Location in Revised Manuscript
1	Please provide stronger evidence demonstrating the advantage of the proposed model over the Linear Regression baseline, as the reported performance differences are minimal.	We provide transparent evidence through five-seed evaluation and DM significance testing. The Hybrid Residual model achieves near-parity (mean DM $p = 0.20$, not significant). Contribution honestly reframed as decomposed, auditable architecture. Multi-site evaluation across 10 Indian states confirms this; Gujarat (GJ) is the only state with consistent significant advantage ($p < 0.05$ across all 5 seeds).	Abstract, Section VI, Section VIII-H, Conclusion

Comment No.	Reviewer Comment	Response	Location in Revised Manuscript
2	Include statistical significance analysis (e.g., Wilcoxon signed-rank test) to support the reported improvements.	Diebold–Mariano (DM) test conducted—the standard significance test for forecast comparison. Mean DM statistic = -0.39 , $p = 0.20$ (not significant at $\alpha = 0.05$). Only 1/5 seeds achieves significance. DM chosen over Wilcoxon because it accounts for serial correlation in prediction errors.	Section VI-D, Table VIII
3	Strengthen the blockchain contribution by providing quantitative evaluation and clearer justification of its necessity within the proposed framework.	Fully evaluated Solidity prototype with 12-module testing suite (62 tests, all passing). Quantitative: gas costs, 2.47 ms latency, 709 TPS, 2.24:1 compression, block finality at 1s/3s/5s intervals, 100% hash integrity. Necessity justified via mechanism comparison table showing only our PGS-gated system provides all four capabilities.	Section VII-G, Tables XII–XVII, Table XI
4	Compare the proposed method with recent state-of-the-art forecasting models to better establish its competitiveness.	Comparison with 7 models: LR, Random Forest, LSTM, GRU, BiLSTM, XGBoost, and 3 EA-BiLSTM variants. Multi-site evaluation across 10 states. The Hybrid Residual model outperforms all standalone deep-learning models (RMSE 0.007187 vs. LSTM 0.015304, BiLSTM 0.021148). Key insight: residual decomposition outperforms end-to-end deep learning on structured datasets.	Section V-D, Table V, Section VIII-H
5	Revise the manuscript for clarity, conciseness, and language quality, as several sections contain repetition and overly detailed explanations.	Manuscript significantly revised: redundant paragraphs removed, technical language standardized, model naming unified to “Hybrid Residual” throughout.	Throughout manuscript

Additional Improvements Beyond Reviewer Suggestions

In addition to addressing all reviewer comments, the following improvements were made:

No.	Improvement	Description	Location
1	Multi-Site Generalizability	Pipeline applied to 9 additional Indian states with 5-seed DM testing. Across 10 states, Hybrid Residual achieves comparable performance to LR (mean RMSE: 0.008626 vs. 0.007787).	Section VIII-H, Table XIX
2	Comprehensive Blockchain Testing Suite	12-module, 62-test-case automated suite covering hash integrity, chain linkage, PGS gating, gas economics, latency, throughput, storage, block finality, and SAST security. All tests pass.	Section VII-G, Tables XII–XVII
3	Root-Cause Analysis of BiLSTM Failure	Formal diagnosis of why 3 standalone BiLSTM architectures failed to outperform LR, with 4 mechanistic hypotheses and confidence assessments.	Section VIII-B, Table XX
4	12 New IEEE References	Added 12 references from IEEE Access, IEEE Trans. Smart Grid, and IEEE Trans. Syst. Man Cybern., covering attention-BiLSTM, blockchain energy trading, IPFS, residual learning, and DL surveys.	References
5	Naming Standardization	All model references unified to “Hybrid Residual” across tables, figures, text, and captions.	Throughout
6	Blockchain Scope Clarification	Explicit statements added that confirmation times are local Hardhat automine; production-network evaluation remains future work.	Section VII-G
7	Abstract Rewrite	Rewritten to reflect near-parity with LR, remove misleading claims, and include blockchain evaluation metrics.	Abstract